Visualization of Molecular Electrostatic Potentials

Two reproducible pipelines from Turbomole grid output to publication figures

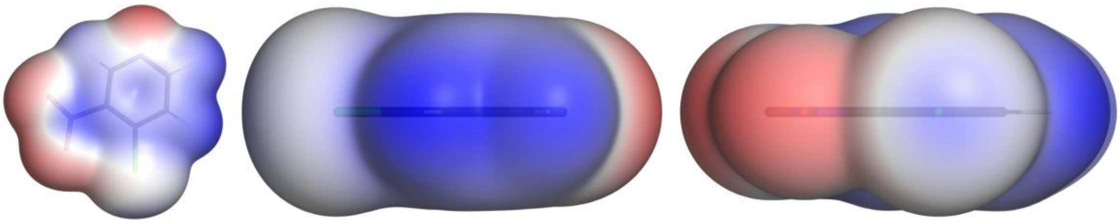

Figure 1: 4-Chloro-3-nitropyridine at ρ = 0.001 a.u. Left: the π face. Centre: the in-plane profile. Right: the view along the C–Cl axis, with the blue σ-hole cap in the middle of the chlorine (shifted by 6.4°) Colour scale ±0.0600 a.u., PyMol output.

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1 Introduction

1.1 Electrostatics as a recognition principle

Non-covalent recognition between a ligand and its target, to a first approximation, is electrostatic. The molecular electrostatic potential on the surface of a molecule is therefore one of the more directly interpretable quantities in structure-based design: it says where a molecule is electron-rich and where it is electron-poor, in units that can be compared between molecules.

The halogen bond is the case where this matters visibly. A covalently bound halogen carries an anisotropic potential: negative in a belt perpendicular to the C–X bond and positive in a cap on its extension. That positive cap referred to as the **σ-hole** is what allows a chlorine, bromine or iodine to act as an electrophile towards a carbonyl oxygen, a backbone nitrogen or a ring π system, forming non-covalent bonds.

1.2 Aim and scope

The goal of the project is to implement a reproducible workflow to turn Turbomole pointval input data into a standard set of ESP figures and a live scene within the designated visualization tool. As used visualization tools, as explained in the */docs/SoftwareEvaluation.pdf* file, PyMol and VMD were chosen. Additionally, the electrostatic potential for the sigma-hole at the halogen is computed.

1.3 The molecular electrostatic potential

The electrostatic potential V(r) at a point r is the energy a unit positive test charge would feel there. It has two contributions of opposite sign: the nuclei, which are positive and concentrated and the electrons, which are negative and diffuse. Where the electron density outweighs the nuclear contribution, the potential is negative; where it does not, there is a positive electrostatic potential.

Two grids are therefore needed and they answer different questions. The density ρ determines **where the surface lies** (shape). The potential V determines **what colour a point on it gets** (the property). This is why the workflow always handles a pair of cube files and why a viewer must be able to colour the isosurface of one dataset by the values of a second (requirement R2 in the *SoftwareEvaluation.pdf* file).

1.4 Why ρ = 0.001 a.u.

The surface used throughout the implementation is the electron-density isosurface at ρ = 0.001 a.u. This is the convention introduced by Bader and adopted by Politzer and Murray: it encloses roughly 96–97 % of the electronic charge and is linearly interpolated by PyMol and VMD itself. Both VMD and PyMol generate it by marching cubes on the density grid and the potential is mapped onto the resulting vertices afterwards. The isovalue is not fixed, it can be changed with the --iso option described in the respective *README.md* files. §4.1 shows what that costs: It is the parameter that moves the reported σ-hole most.

1.5 $V_{S,min}$ and $V_{S,max}$

The two conventional descriptors are the most negative and most positive values of the potential on the molecular surface: $V_{S,min}$ and $V_{S,max}$. They are global extrema over the whole molecular surface and are important for evaluating the scale of the spheres colouring.

1.6 The σ-hole

On an aryl halide the global maximum of the surface potential does not sit on the halogen. It sits on the ring hydrogens. The three halogen-benzenes make the point cleanly, all measured on the same 251^3 grid at $\rho = 0.001$ a.u.:

	chlorobenzene	bromobenzene	iodobenzene
σ-hole (cap on the C–X axis)	+0.00786 a.u. +4.9 kcal/(mol·e)	+0.01629 a.u. +10.2	+0.02559 a.u. +16.1
ring hydrogens = global V_S,max	+0.03129 a.u. +19.6	+0.03154 a.u. +19.8	+0.03168 a.u. +19.9
halogen belt = global V_S,min	−0.01896 a.u. −11.9	−0.01882 a.u. −11.8	−0.01685 a.u. −10.6

Table 1: The three global maxima agree to within 1.3 %, because all three are the same aromatic C–H. The σ-holes span a factor of 3.3 in the expected order Cl < Br < I.

The consequence for the workflow is that the σ-hole must be located as a **local** maximum in a cone around the extended C–X axis. Its counterpart, the halogen belt, is the minimum in a band perpendicular to that axis. Together the two numbers describe the anisotropy that makes a halogen bond directional: positive ahead, negative to the side.

2 Data

2.1 Quantum-chemical input

All grids were produced with Turbomole data and each molecule comes as a pair of *pointval* files: *td.xyz* holding the electron density and *tp.xyz* holding the electrostatic potential, on the same grid. Additionally a structure file in `.mol`, `.sdf` or `.xyz` format is provided and crucial for the workflow.

2.2 The molecule set

The provided data for the pipelines, consisted of one set of halogen-benzenes and one set of pyridine derivates, it follows the two questions the report asks of the chemistry. The halobenzenes vary the halogen with everything else held constant, which isolates the polarizability trend. The pyridines vary the substituent with the ring held constant, which isolates the electronic effect on the lone pair. Additionally with the help of the */tools/CreateTpTdFromSmiles.py* converting tool, Triazolam, a benzodiazepine that carries 2 halogens in a smaller grid (see table 3) was generated and used as a test object. Several other molecules have been created and tested with the pipelines, but aren't reported, as the focus should stay on the pyridines and halobenzens.

Folder	Molecule	Role in the set
Pyridine	pyridine	reference point for the substituent series
Me-Pyr	4-methylpyridine	electron-donating substituent
CN-Pyr	4-cyanopyridine	electron-withdrawing substituent
NO2-Pyr	4-nitropyridine	strongly electron-withdrawing substituent
I-Pyr	3-iodopyridine	halogen on an electron-poor ring
Cl-NO2-Pyr	4-chloro-3-nitropyridine	halogen with an adjacent nitro group
brombenzol	bromobenzene	halobenzene series
triazolam	triazolam	testing purpose only

Table 2: The molecules within the /results folder.

3 Results

3.1 Surface ESP values

All values are calculated at an isovalue of ρ = 0.001 a.u. Atom labels are element symbol plus the 1-based index into their respective structure file. σ-hole and belt is calculated with the ray method (see 1.2.2 in *PythonElaboration.pdf*), the angle is the deviation of the σ-hole maximum from the C–X axis. 4-Cl-3-NO_2-Pyridine shows a bent angle of 6.4°, which is influenced by the neighbouring NO_2 group, as indicated in figure 1. Triazolam shows the benefits of the PyMol implementation, as it comes with two halogens, the halogen with the bigger esp value is taken for the subsequent workflow. This feature is outsourced in the VMD implementation, to decrease the runtime.

Molecule	Grid	Shell points	V_S,min (a.u.)	V_S,max (a.u.)	σ-hole (a.u.)	kcal/(mol·e)	Angle	Belt (a.u.)
pyridine	251^3	12447	−0.06500 (N7)	+0.03959 (H6)	—	—	—	—
4-methylpyridine	251^3	14641	−0.06779 (N10)	+0.03647 (H12)	—	—	—	—
4-cyanopyridine	251^3	14505	−0.05927 (N3)	+0.05025 (H8)	—	—	—	—
4-nitropyridine	251^3	14452	−0.04712 (N2)	+0.04487 (H12)	—	—	—	—
3-iodopyridine	251^3	16843	−0.05638 (N8)	+0.04594 (H10)	+0.03441 (I7)	+21.6	0.0°	−0.01147
4-chloro-3-nitro-pyridine	251^3	16332	−0.04545 (O8)	+0.05825 (H12)	+0.02508 (Cl10)	+15.7	6.4°	−0.03091
chlorobenzene	251^3	35592	−0.01896 (Cl12)	+0.03129 (H9)	+0.00786 (Cl12)	+4.9	0.0°	−0.01896
bromobenzene	251^3	38010	−0.01882 (Br12)	+0.03154 (H7)	+0.01629 (Br12)	+10.2	0.0°	−0.01882
iodobenzene	251^3	41631	−0.01685 (I12)	+0.03168 (H5)	+0.02559 (I12)	+16.1	0.0°	−0.01685
triazolam	132×101 ×106 self-generated	8146	−0.08429 (N4)	+0.05432 (H27)	+0.01709 (Cl21) +0.01440 (Cl11)	+10.7 +9.0	3.8° 1.3°	−0.01836 −0.01095

Table 3: Complete result set, from the batch run of 31 August 2026. Both pipelines produce the V_S columns; the σ-hole, angle and belt columns come from the PyMol implementation.

3.2 The halobenzene series

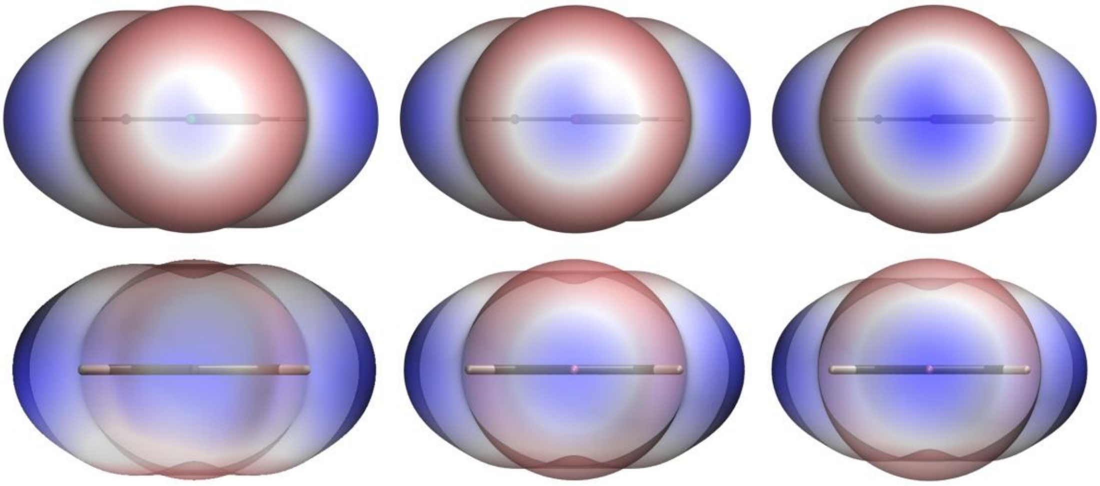

Figure 2: The view along the C–X axis for chlorobenzene, bromobenzene and iodobenzene. PyMol above, VMD below, identical data, identical isovalue, identical ±0.0350 a.u. scale throughout. The σ-hole grows from a pale centre through a defined blue disc to a large saturated cap.

The series behaves exactly as the polarizability argument predicts: +4.9, +10.2 and +16.1 kcal/(mol·e) for chlorine, bromine and iodine, a factor of 3.3 across the three. The heavier the halogen, the more its electron density is drawn towards the bond and the more of the nuclear charge is exposed on the far side, expanding the potential at the sigma-hole.

The belt moves in the opposite direction and by much less: −0.01896, −0.01882 and −0.01685 a.u. becoming slightly less negative as the halogen gets heavier and more polarizable. The anisotropy of the halogen therefore increases in both senses at once, but the σ-hole is by far more sensitive.

3.3 The pyridine series

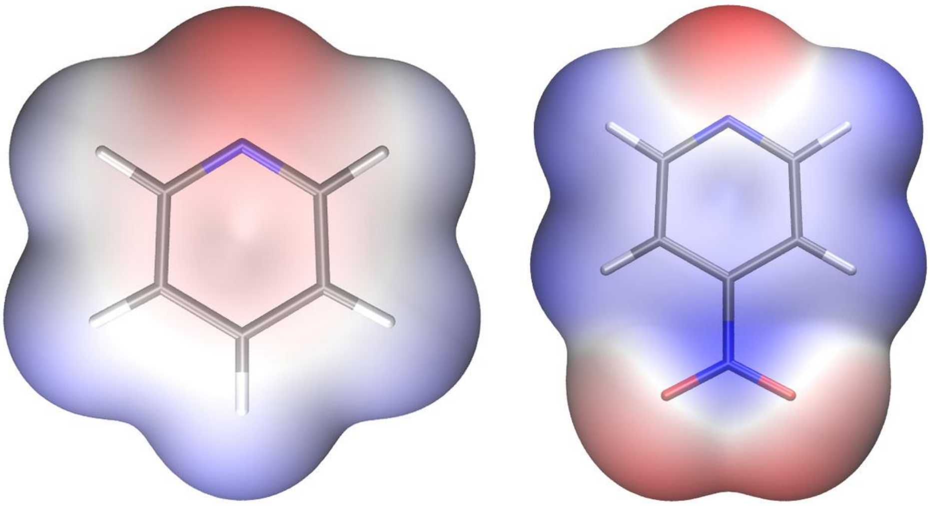

Figure 3: The π face of pyridine (left, scale ±0.0700 a.u.) and 4-nitropyridine (right, ±0.0500 a.u.). The ring nitrogen is the deep red lone pair at the top. The nitro group of the right-hand molecule pulls density out of the ring, and the lone pair becomes measurably less negative. Rendered with the VMD implementation.

In five of the six pyridines $V_{S,min}$ sits on the lone pair of the nitrogen atom. The substituent shifts it in the expected direction and by a large amount: methyl −0.06779, unsubstituted pyridine −0.06500, cyano

−0.05927, nitro −0.04712 a.u. That is a span of 0.0207 a.u., 54 kJ/(mol·e), or 31 % of the unsubstituted value, from changing one substituent at the far end of the ring.

The exception is 4-chloro-3-nitropyridine, where $V_{S,min}$ has moved off the ring nitrogen onto a nitro oxygen (O8). With two electron-withdrawing groups the lone pair is depleted enough that the nitro oxygens become the most negative site on the molecule.

3-Iodopyridine reaches a σ-hole value of +0.03441 a.u. (+21.6 kcal/(mol·e)) against +0.02559 (+16.1) for iodobenzene: the same C–I bond on an electron-poorer ring is 34 % stronger. Within 4-Chloro-3-nitropyridine the chlorine reaches +0.02508 against +0.00786 on chlorobenzene, more than three times as much, so that a chlorine in the right electronic environment produces a σ-hole comparable to an iodine in a neutral one. For a medicinal chemist that is the actionable result: t**he ring matters as much as the halogen**.

It is also the only σ-hole in the set that is not on the bond axis. Its maximum sits 6.4° off, pushed there by the neighbouring nitro group, every other halogen here reports its maximum value at the sigma hole at 0.0°.

3.4 Comparison with published values

The values above are internally consistent, but to prove reliability, they are compared to the literature. The halobenzenes are the right molecules for an external check, because they are among the most frequently computed σ-hole systems in the literature.

Molecule	This work	Riley 2011 B3PW91/6-311G*	Devore 2024 M06-2X/aug-cc-pVDZ(-PP)	Heidrich 2019	Literature range
chlorobenzene	+4.9	+5.3	+4.1	+5.02	4.1 – 5.3
bromobenzene	+10.2	+12.2	+10.9	+10.04	10.0 – 12.2
iodobenzene	+16.1	+17.3	+16.1	+15.7	15.7 – 17.3

Table 4: σ-hole V_S,max in kcal/(mol·e), all on the ρ = 0.001 a.u. surface. Sources in §8. All values are in **kcal/(mol·e)**

All three values fall inside the published range. The spread among the literature sources themselves is roughly ±10–20 % and the values obtained here sit comfortably within that band. The Heidrich values were published in atomic units and converted here. Its values are chlorobenzene 0.008 a.u., bromobenzene 0.016 and iodobenzene 0.025. The computed values at the trilinear interpolation step of the sigma holes of our implementation are: chlorobenzene 0.00786, bromobenzene 0.01629 and iodobenzene 0.02559 a.u. This range seems satisfying and serves as confirmation for the calculation step.

Heidrich et al. report two different Vmax quantities and only one of them is comparable here: the σ-hole magnitudes attached to their fragment library are values *predicted* at an isodensity of 0.020 a.u. by a machine-learned SVM model (VmaxPred), whereas the halobenzene reference values quoted above are *computed* Vmax values on the classical ρ = 0.001 a.u. surface.

The isosurface matters far more than the method and this is the trap the comparison had to avoid. The same three molecules computed on the ρ = 0.002 a.u. surface give +10.1, +18.7 and +25.5 kcal/(mol·e) (Ibrahim et al., 2022) between 1.5 and 2.2 times the 0.001 values, our own value on the same surface is +16.6 for bromobenzene, the remaining difference is the method. Any quoted σ-hole is meaningless without its isosurface, which is the same conclusion the parameter study reaches from the other direction (§4.1).

3.5 The image sets

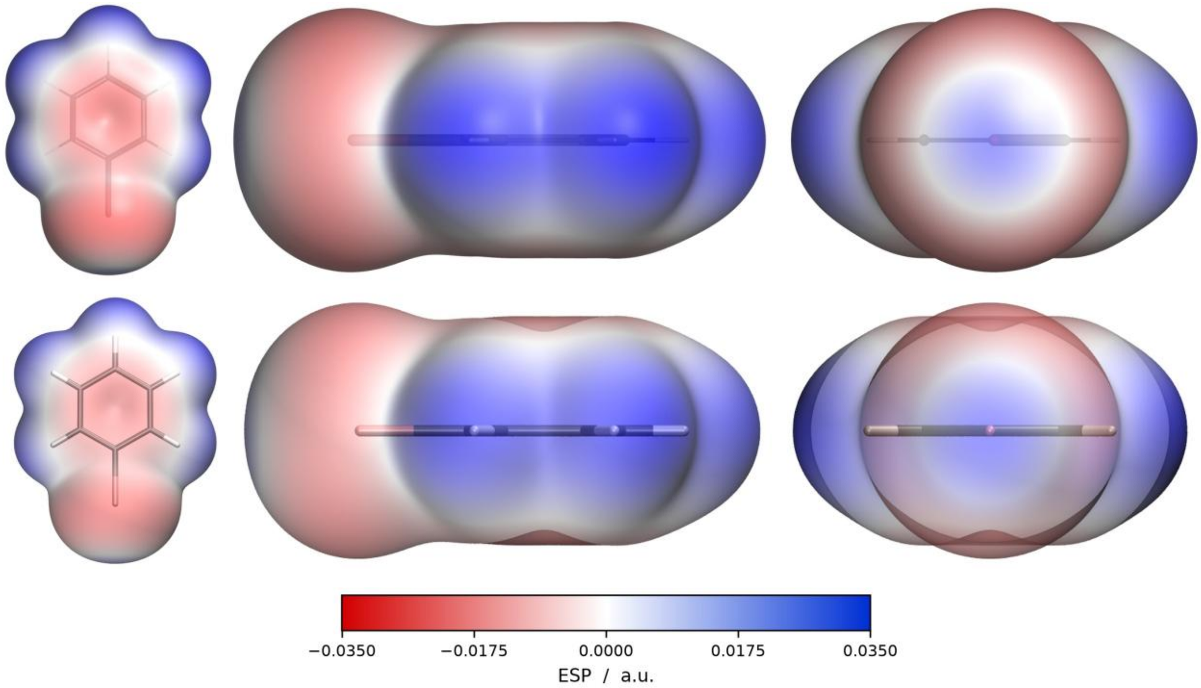

Figure 4: Bromobenzene, the three standard views: π face, in-plane profile, and the axial view down the C–Br bond. PyMol above, VMD below, both at ±0.0350 a.u. The potential maps agree; the differences are in the rendering, VMD draws the more readable skeleton, PyMol the more saturated surface.

For each molecule three views plus a colour bar and a settings file are delivered. The three views are computed from the geometry: the ring normal comes from the principal axis of smallest extent over the heavy atoms and the axial view follows the true, unprojected C–X axis.

3.6 Pipeline comparison

Molecule	V_S,min PyMol	VMD	V_S,max PyMol	VMD	Shell points PyMol / VMD
pyridine	−0.06500	−0.06500	+0.03959	+0.03959	12447 / 12447
4-methylpyridine	−0.06779	−0.06779	+0.03647	+0.03647	14641 / 14640
4-cyanopyridine	−0.05927	−0.05927	+0.05025	+0.05025	14505 / 14504
4-nitropyridine	−0.04712	−0.04712	+0.04487	+0.04487	14452 / 14452
3-iodopyridine	−0.05638	−0.05638	+0.04594	+0.04594	16843 / 16840
4-chloro-3-nitropyridine	−0.04545	−0.04545	+0.05825	+0.05825	16332 / 16330
chlorobenzene	−0.01896	−0.01896	+0.03129	+0.03129	35592 / 35592
bromobenzene	−0.01882	−0.01882	+0.03154	+0.03154	38010 / 38008
iodobenzene	−0.01685	−0.01685	+0.03168	+0.03168	41631 / 41629
triazolam	−0.08429	−0.08429	+0.05432	+0.05432	8146 / 8145

Table 5: All twenty surface extrema are identical to every printed digit and so are all ten colour scales.

The agreement extends to the σ-hole, it was calculated in the same manner, with the help of trilinear interpolation: render_esp.py inside the PyMol render pass and tools/SigmaHoleCalc.py in the VMD repository.

Molecule	Halogen	σ-hole PyMol (a.u.)	σ-hole VMD (a.u.)	kcal/(mol·e)
chlorobenzene	Cl12	+0.00786	+0.00786	+4.9
bromobenzene	Br12	+0.01629	+0.01629	+10.2
iodobenzene	I12	+0.02559	+0.02559	+16.1
3-iodopyridine	I7	+0.03441	+0.03441	+21.6
4-chloro-3-nitropyridine	Cl10	+0.02508	+0.02508	+15.7
triazolam	Cl21	+0.01709	+0.01709	+10.7
triazolam	Cl11	+0.01440	+0.01440	+9.0

Table 6: The σ-hole of every halogen in the set, computed twice on the same cube files, once inside the PyMol render pass and once by the standalone tool in the VMD repository. Identical to five decimals throughout, including both chlorines of triazolam, which the two find and rank separately.

The only measurable disagreement is in the shell point counts, which differ by 0 to 3 out of 8 000 to 42 000. The cause is the data type: one converter holds the grid in float32 and the other in float64, so a handful of points sitting exactly on the edge of the ρ = iso ± 12 % band fall on different sides of it. No extremum is affected, because those edge points are never the extreme ones.

What this establishes. The two pipelines see the same surface, with the same values, on the same scale. Every difference between the two rows of Figures 2, 4, 5 and 6 is therefore attributable to the viewer, to how it triangulates the isosurface, how it interpolates the potential onto the vertices, and how it composites transparency. The numbers are viewer-independent; the images are not.

Two further image pairs show what the viewer does contribute and they separate the two mechanisms that cause it.

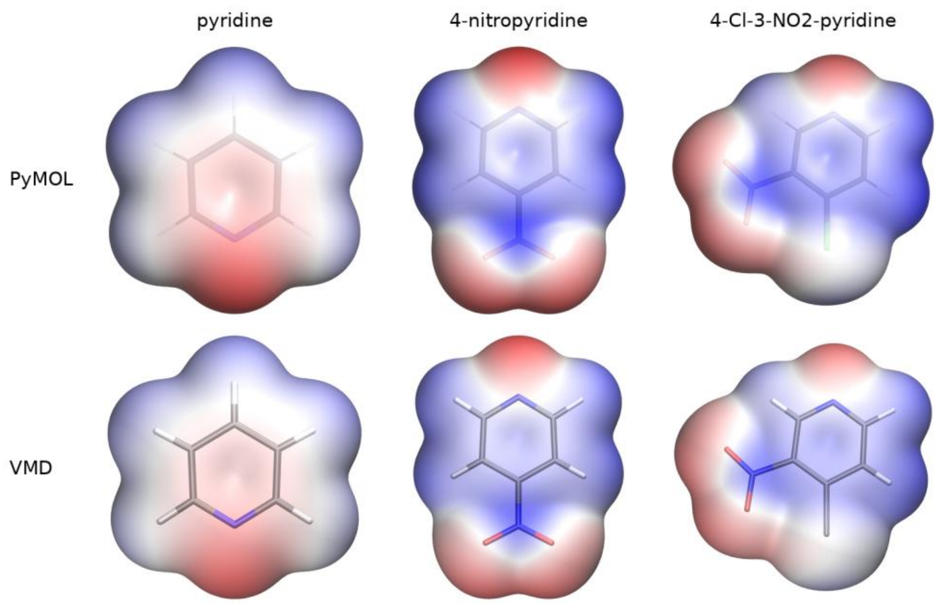

Figure 5: The π face of pyridine, 4-nitropyridine and 4-chloro-3-nitropyridine, PyMol above and VMD below. Scales ±0.0700, ±0.0500 and ±0.0600 a.u. respectively, so the rows may be read against each other, but the columns may not.

In this view the balance tips towards VMD. The potential maps agree closely, ring nitrogen and nitro oxygens red, ring face blue, the same gradients, but VMD draws the more legible skeleton. Its sticks carry element colours through the surface, so nitrogen, oxygen and the hydrogens stay identifiable, where PyMol's are a faint grey outline at the same transparency setting. For a figure whose job is to show where on a molecule a potential sits, that is an advantage, and it is the one place in this comparison where the second pipeline produces the better picture.

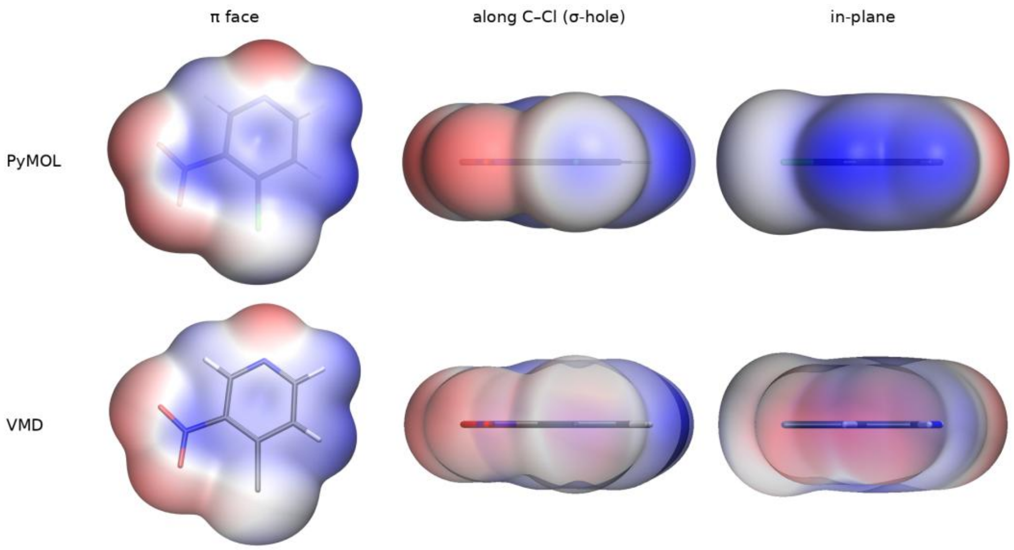

Figure 6: 4-Chloro-3-nitropyridine, the three standard views, PyMol above and VMD below, ±0.0600 a.u.

This molecule is the clean test. The π face repeats the finding of Figure 5: VMD's coloured skeleton makes the substitution pattern more distinguishable. The axial view repeats the finding of Figure 2: PyMol resolves the pale σ-hole on the chlorine inside its red belt cleaner, VMD flattens the same surface into an even wash. Both are full-quality renders of identical data, the difference results from the transparency weighting. Changing the opacity value within the VMD implementation to 0.8 results in the following pictures. The molecules skeleton, becomes washed out, but the axial view, showing the sigma hole becomes more precise, comparable with the PyMol picture:

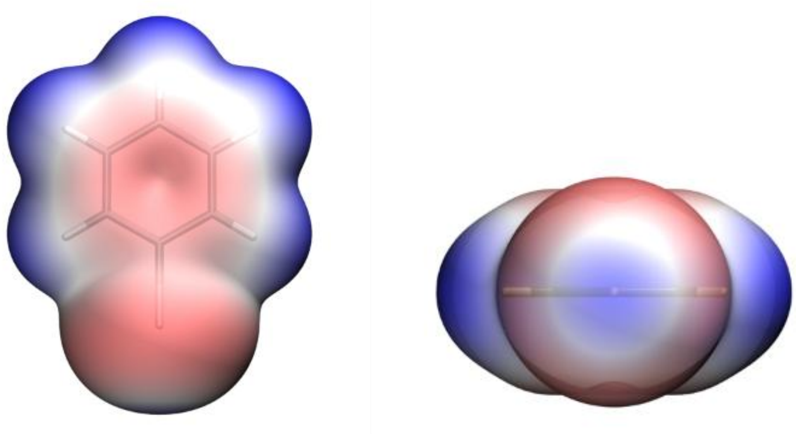

Figure 7: bromobenzene, ±0.035 a.u. pictures generated with the VMD pipeline, with an --opacity value of 0.8

As stated in §5.3 the two tools measure transparency in different manners. The VMD via an opacity and the PyMol tool via a transparency value. All images have been created in VMD with an opacity value of 0.5 and in PyMol with a transparency value of 0.15, Figure 7 and Figure 8 show the results with a different transparency value set, increasing the opacity in the VMD implementation, makes the Sigma-hole better identifiable and increasing the transparency value in the PyMol implementation makes the sticks identifiable. The mere differences between the two image sets can be negated with changing the corresponding transparency value.

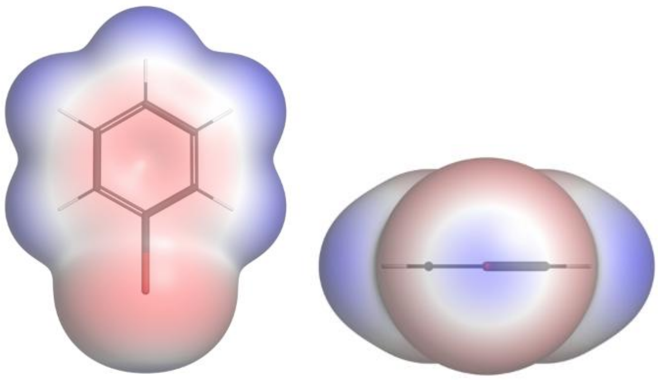

Figure 8: bromobenzene, ±0.035 a.u. pictures generated with the PyMol pipeline, with an --transparency value of 0.4

3.7 The rainbow ramp

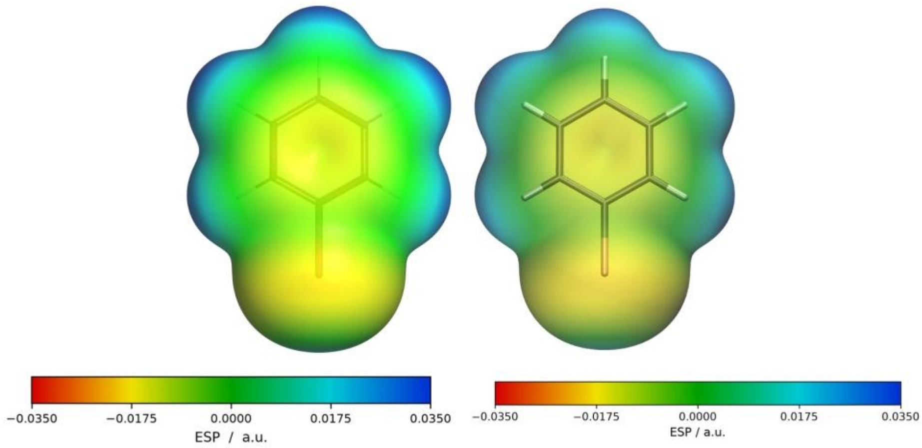

Figure 9: Bromobenzene with the rainbow ramp, PyMol left and VMD right, each pipeline generates its own colour bar. Scale ±0.0350 a.u.

Both pipelines offer a five-stop ramp: red-yellow-green-cyan-blue as an alternative to red-white-blue, and both write it as a separate image set. The mechanisms could hardly be more different: PyMol creates a ramp object with five explicit anchor levels, while VMD has no ramp object at all and the effect has to be produced by rewriting 1024 entries of its global colour table. That the two produce the same colours at the same places is the useful negative result of the comparison.

The asymmetry is worth one more sentence, because it is the sharpest illustration of how little the two share. PyMol's ramp_new takes the five colours and the five values and interpolates between them internally, in one call. VMD has no ramp object at all, only a fixed 1024-entry colour table, so the generated scene walks all 1024 entries, works out which of the four segments between the five anchor colours each entry falls in and mixes R, G and B linearly between the two neighbouring anchors. The Python side only converts the five hex colours into RGB triples; the interpolation itself happens in Tcl, inside the scene file. Both routes put the same colour at the same value.

When to use which. Red–white–blue spends almost no colour resolution near zero, so everything weakly polar comes out white; the rainbow resolves exactly that region. The price is that a rainbow is not perceptually uniform. The eye reads the yellow–green and green–cyan transitions as edges even though the potential changes smoothly. For a quantitative claim, red–white–blue remains the more honest depiction. The convention is deliberately not inverted in either ramp: red stays negative and blue stays positive, so the two sets can be laid side by side.

4 Parameter selection

This section collects the measurements and it is the part of the report that generalises beyond this dataset. **The short version: the isovalue is the only parameter that changes the physics.**

4.1 Isovalue

Bromobenzene on the full 251^3 grid at 0.12 Bohr spacing, everything else at defaults:

ρ / a.u.	Shell points	V_S,min	V_S,max	σ-hole	kcal/(mol·e)	Belt
0.0005	43009	−0.0175	+0.0248	+0.0098	+6.2	−0.0175
0.0008	39451	−0.0185	+0.0292	+0.0139	+8.7	−0.0185
0.0010	38010	−0.0188	+0.0315	+0.0163	+10.2	−0.0188
0.0015	35283	−0.0191	+0.0377	+0.0217	+13.6	−0.0191
0.0020	33771	−0.0191	+0.0424	+0.0265	+16.6	−0.0191
0.0040	30320	−0.0173	+0.0615	+0.0434	+27.2	−0.0173

Table 7: From 0.0005 to 0.004 the σ-hole grows by a factor of 4.4.

A larger isovalue means a surface closer to the nuclei, where less of the positive nuclear contribution has been screened by the electrons. But the isovalue does not simply rescale the surface: over the same range the global maximum on the ring hydrogens grows by a factor of 2.5 while the belt around the bromine barely moves and then reverses. Different features respond differently, which is precisely why a σ-hole value quoted without its isovalue is meaningless and why two values computed at different isovalues cannot be compared at all.

The --iso option exists to reproduce someone else's choice, not to improve on 0.001.

4.2 Grid resolution

Same molecule, isovalue fixed at 0.001, the same grid decimated by an increasing stride during converting the *td.xyz* and *tp.xyz* file into *.cube* files, decimating the grid is possible via --stride:

--stride	Grid	Δ / Bohr	Cube size	V_S,min	V_S,max	σ-hole	kcal/(mol·e)
1	251^3	0.12	201.1 MB	−0.0188	+0.0315	+0.0163	+10.2
2	126^3	0.24	25.4 MB	−0.0188	+0.0312	+0.0161	+10.1
3	84^3	0.36	7.5 MB	−0.0188	+0.0312	+0.0160	+10.1
4	63^3	0.48	3.2 MB	−0.0188	+0.0311	+0.0158	+9.9
6	42^3	0.72	0.9 MB	−0.0188	+0.0312	+0.0149	+9.4
8	32^3	0.96	0.4 MB	−0.0187	+0.0296	+0.0135	+8.4

Table 8: Cost falls by a factor of 480 across this table; the σ-hole value accuracy drops by 17 %.

The degradation is smooth and one-sided, because the interpolated density smooths the isosurface outwards and the potential there is less positive. $V_{S,min}$ is essentially immune it is a broad, flat feature while the σ-hole, a sharp peak, is the sensitive one. **Practical rule: stride 2 for creating the images and the interactive scene, stride 1 for numbers that go in a table.** The converter warns above 0.30 Bohr spacing for exactly this reason.

4.3 Evaluation method: rays against grid points

The same six grids, the same isovalue, the same molecule, only the way the σ-hole is read off differs:

Grid	Spacing	Point-based	Ray-based (used)
42^3	0.72 Bohr	+2.1 kcal/(mol·e)	+9.4 kcal/(mol·e)
126^3	0.24 Bohr	+7.9	+10.1
251^3	0.12 Bohr	+11.0	+10.2

Table 9: Bromobenzene. The point-based value spans a factor of five across these grids; the ray-based one varies by 8 %.

Evaluating the sigma-hole in a trilinear interpolated fashion, results in way more accurate results than using the point that is closest to the calculated radius of the sigma-hole.

4.4 σ-hole search parameters

The search itself is governed by six settings: a cone half-angle of 36.9°, 400 rays on a Fibonacci spiral with the first ray exactly on the axis, a 0.02 Bohr step, a radius cut-off at 1.6 × the Bondi van der Waals radius and a belt half-angle of ±69.5° at 1.5 × the mean cap radius. The first two were tuned; the rest follow from the geometry and all six are documented with their reasoning in the code.

4.5 Colour range

The colour range maps numbers to colours. It appears in this section because getting it wrong makes a correct calculation look wrong.

Molecule	Automatic range	Set by
halobenzenes	±0.035	the halogen belt
4-chloro-3-nitropyridine	±0.060	the ring hydrogens
pyridine, 4-methylpyridine	±0.070	the ring nitrogen lone pair
triazolam	±0.085	the triazole nitrogens

Table 10: The automatic range is set by whichever feature of the molecule is most extreme, which is rarely the feature under discussion.

A σ-hole of +0.017 a.u. is 49 % of full blue at ±0.035 and 20 % at ±0.085. Therefore it is clearly visible in the first case and nearly white in the second, from identical data. Triazolam is the cautionary example: its triazole nitrogens push the automatic range out to ±0.085 and a correctly computed σ-hole but it disappears from the picture, due to the sheer scale. **A common scale is required whenever two figures are compared and a common scale is meaningless across data of different provenance.**

4.6 Rendering parameters

None of the rendering settings changes a computed value: transparency 0.15 in PyMol against opacity 0.50 in VMD, a 0.10 Å stick radius, cast shadows off in both, orthoscopic projection and 2000 × 1600 px at 300 dpi in PyMol against 1600 × 1061 in VMD, where the image is capped at window size. Every value that produced a figure is recorded in the *_settings.txt beside it, which is the reproduction record.

5 Robustness and failure modes

A workflow that fails loudly is better than one that fails quietly and every defect described here produced a plausible-looking figure before it was caught. The implementations of these guards are described in PythonElaboration.pdf, this section reports what they were built against and what they cost.

5.1 Structure and grid do not belong together

A molecule folder holds the two grids and a structure file. If it holds more than one structure files that are not simply the same geometry in a second format, the batch driver used to take the first in sort order. Both pipelines now abort the process, name every candidate they found and ask for the folder to be cleaned up.

When a wrong structure file is inputted and the .cube files are created with the wrong structure file, the molecule floats beside its own isosurface within the created images/scene.

The guard is one line of physics. Where a nucleus sits the electron density is enormous, where none sits it is near zero. The check looks up the density at every heavy nucleus and takes the median of the largest value in the 3×3×3 neighbourhood of each:

Grid spacing	Median ρ at the heavy nuclei	Interpretation
0.12 Bohr	64	full resolution
0.25 Bohr	27	typical production run
0.60 Bohr	5	the decimated reference dataset
—	0.03	the broken case: wrong structure

Table 11: The threshold is set at 1.0, a factor of five below the coarsest healthy case and thirty above the broken one. A coarse grid depresses the value because the nucleus then sits further from the nearest grid point and the density peak falls off steeply.

In a batch the two failures are treated differently. Ambiguous structure files stop the whole run, because nothing can be chosen. A structure/grid mismatch is local to one folder: at several minutes per molecule, failing a ten-molecule run over one bad folder would be waste, so the molecule is skipped, named again at the end. Visible to whatever called the script, without the other nine image sets being lost.

5.2 The ray leaving its own halogen

Triazolam exposed a second defect and this one was not about having two halogens. It had simply never been triggered by flat molecules. The ray method originally took the *outermost* crossing of ρ = 0.001. In a folded molecule the cone around one C–X axis does not point into empty space but at another part of the molecule, so the ray passed through the halogen's own surface, through the gap and measured the surface of the methyl group on the triazole ring beyond it.

	Before	After
Cl21 σ-hole	+18.8 kcal/(mol·e), 29.0° off the axis	+10.7, 3.8°
Cl21 belt	−42.0 (in fact the triazole nitrogens)	−11.5

Table 12: +18.8 for an aryl chloride was the clue: chlorobenzene gives +4.9, and no substituent pattern triples that.

Two changes fixed it, and both are now defaults:

– The ray takes the innermost downward crossing. It starts deep inside the halogen's own density, so the first drop through the isovalue is that halogen's own surface.

– Both the ray search and the grid-point cap are cut off at 1.6 × the Bondi van der Waals radius of the halogen.

Single-halogen molecules are unaffected. Bromobenzene stays at +0.01629 a.u. and −0.01882 for the belt, identical to every digit before and after. **The lesson generalises: three flat halobenzenes cannot test a method that assumes the space beyond the halogen is empty.** The same molecule also revealed that the axial view was being computed from the C–X axis projected into the best-fit plane, which for a non-planar molecule looked 42.9° past the σ-hole. Both defects were invisible on the model compounds and obvious on a real drug molecule, resolved within the implementation.

5.3 Renderer behaviour

Three properties of the VMD renderer had to be worked around and they are documented because they affect what the images can be used for.

Transparency does not translate between the programs and the two settings are not even on the same scale. PyMol is set by transparency, where 0 is opaque and the pipeline uses 0.15; VMD is set by the opacity of its Transparent material, where 1 is opaque and the pipeline uses 0.50. PyMol's 0.15 transparency is therefore an opacity of a = 0.85 and a value copied from one program into the other means close to its opposite.

Converting is not enough either. The isosurface is closed, so a line of sight enters at the front and leaves at the back and at opacity a only $(1 - a)^2$ of what lies behind survives: 25 % at VMD's a = 0.50, and 2 % at PyMol's a = 0.85. By that arithmetic PyMol's images should be solid and they are not. The stick model shows through clearly, because transparency_mode 2 treats the back faces of the surface differently from a naive two-layer product. VMD applies no such correction and empirically needs roughly a = 0.80, which is 4 % transmission, before the axial view gives a comparable impression. The two programs do not merely label the setting differently, they even weight it differently.

One setting cannot serve all three views. The axial view crosses the most layers of surface and would need the highest opacity to read like the others; at that value the π and in-plane views, turn so opaque that their molecule skeleton colour can no longer be read. The pipeline keeps a single value per run, so that the three images of one molecule stay mutually comparable and the cost is paid uniformly across every VMD panel: the red belt separates less clearly from the blue cap.

The ray tracer fails unpredictably on stacked transparency. In the axial view every line of sight crosses many transparent layers and TachyonInternal (the VMD renderer) aborts there. On Windows with exit code 0xC0000374, heap corruption and no error message. The same nine molecules were rendered twice from identical data with identical settings: nine of twenty-seven views fell back to a window capture in the first run and seven in the second, with the set almost completely exchanged. Iodobenzene lost all three views in the first run and none in the second; bromobenzene lost one and then two. **The failure is therefore not a property of the molecule and it is not reproducible.** As a result the VMD pipeline renders in passes (§2.2.5 in *PythonElaboration.pdf*) and keeps whatever each pass produced, recording per view which pass produced it and writing it in the settings.txt. PyMol’s ray tracer rendered all twenty-seven without a single failure.

Image size is capped by the screen. TachyonInternal renders at window size and Windows clamps the window to the display, so a request for 1600 × 1280 yields 1600 × 1061 on a 1080-line screen. The VMD image sets are correspondingly smaller than the PyMol ones.

6 Discussion

6.1 What is viewer-dependent and what is not

The separation established in §3.6 is clean and as far as this dataset goes, complete. Surface extrema, colour ranges, σ-holes and belts are properties of the data and the convention, not of the viewer: the two pipelines run the same raw Turbomole files through two separately written conversion chains, their own unit handling, index reordering, cube writing and cube reading and they arrive at the same numbers. What this does *not* establish is that the method is right, because the evaluation itself is the same algorithm in both projects; that limitation is stated in §6.3.

Everything that differs between the two rows of Figures 2, 4, 5 and 6, the saturation of the surface, the visibility of the skeleton, the apparent size of the molecule, is the viewer.

That has a practical consequence for reading the literature. A published ESP figure carries no information about the program that made it and it also needs none. What it must carry is the isovalue and the colour range, because those are what the parameter study in §4 points out actually matters. A figure without a labelled colour bar cannot be read at all, this workflow ships one with every image for that reason.

6.2 Which pipeline for what

- For quantities, use the PyMol pipeline. Not because its V_S values are better, they are identical, but because it is the one that locates the σ-hole and additionally names the atom carrying each extremum and handles several halogens per molecule.
- For the π face, either. The maps agree and VMD draws the more readable skeleton, if you wanna use the PyMol pipeline throughout, as stated, the PyMol transparency value can be increased, so the skeleton in the π view becomes visible (Figure 8).
- For the axial view, PyMol. This is the view that carries the σ-hole argument and the one VMD renders worst. Partly through the transparency mechanics, partly because it is the view its ray tracer is most likely to abandon (§5.3). If you want to use the VMD pipeline throughout, the opacity value for the axial view can be changed and as seen in figure 7, the image carries a good visible σ-hole (see 3.2 *VMD_esp_visualization/README.md*).

6.3 Limitations

- The σ-hole tool in the VMD repository is a copy of the PyMol routine, not an independent derivation. Its self test proves the two copies are still the same code; it does not independently confirm the method, the comparison with literature values, does.
- Two grid geometries. The pyridines were computed on a wider box at coarser spacing than the halobenzenes. Table 8 indicates the effect on V_S is negligible at these spacings, but the two groups are not strictly a single dataset.
- Avogadro 2 and GaussView 6 were assessed from documentation only (see *SoftwareEvaluation.pdf*).
- The delivered image sets and the parameter study were produced on the same data but at different times; the tables in §4 come from the bromobenzene study and were not repeated for every molecule.

7 Outlook

7.1 Electrostatic complementarity

The natural extension is from one molecule to an interface: comparing the potential on a ligand surface with the potential the protein presents at the same place, so that a red patch facing a blue one scores as complementary. Several components of this workflow transfer directly: the cube handling, the shell band, the surface statistics and the rendering. Three things would have to be added: an electrostatics model for the protein, which is a different computational problem (Poisson–Boltzmann, or partial charges with a distance-dependent dielectric); a quantitative definition of what “complementary” means as a number; and a validation case for which the answer is already known. The last of these is the hard one and it is why this was scoped out rather than attempted (see *Electrostatic Complementarity as a Fast and Effective Tool to Optimize Binding and Selectivity of Protein−Ligand Complexes*).

7.2 A third viewer

ChimeraX would take the evaluation from two data points to three at low marginal cost: the cube files and the analysis already exist, only a scene script and an image set would be new. On the evidence of §3.6 the numbers would agree; the open question is whether its OpenGL-only rendering produces a comparable axial view, which is the one view where the two existing pipelines genuinely differ.

7.3 The missing axis

Of the influences examined in §4, four were measured here and only the sensitivity to the quantum-chemical method was taken from the literature. Repeating a single molecule at two levels of theory would close that gap with a number of this project's own, which is the most valuable single addition the work could still receive.

8 References

Method and convention

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Software

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- GaussView 6, Gaussian Inc. gaussian.com/gaussview6

Workflow documentation

- README.md in each repository: installation, execution, complete parameter reference, worked examples, troubleshooting. These are the standard operating procedures and are the authoritative reference for running the pipelines.
- tools/README.md: the test-data generator, its parameters and its limitations.